1

$y'' + 2y' = 0$

$\lambda^2 + 2\lambda = 0$

$\lambda \cdot (\lambda + 2) = 0$

$\lambda_1 = 0 \quad \lambda_2 = -2 \rightarrow y(x) = ae^{0x} + be^{-2x}$

OŘ: $y(x) = a + be^{-2x}, \ x \in \mathbb{R}$

$y'(x) = (-2)be^{-2x}$

$y(0) = 12$:

$12 = a + b$

$y'(0) = 2$:

$2 = (-2)b \Rightarrow -1 = b$

$\rightarrow 12 = a - 1$

$a = 13$

řešení: $y(x) = 13 - e^{-2x}$ pro $x \in \mathbb{R}$

bonus:

$(y(x) \sim a, \ x \rightarrow \infty)$

(2)

$y''' - 2y'' + 4y' - 8y = 0$

a)

$\lambda^3 - 2\lambda^2 + 4\lambda - 8 = 0$

$(\lambda - 2)\cdot(\lambda^2 + 4) = 0$

$\lambda = \{2; 2i; -2i\}$

$\pm 2 \quad \pm 4 \quad \pm 8$

$$\begin{array}{l} \lambda^3 - 2\lambda^2 + 4\lambda - 8 : (\lambda - 2) = \lambda^2 + 4 \\ \underline{-\lambda^3 + 2\lambda^2} \\ \quad 4\lambda - 8 \\ \quad \underline{-4\lambda + 8} \\ \qquad 0 \end{array}$$

OŘ: $y(x) = ae^{2x} + b\sin(2x) + c\cos(2x)$

b) $y(x) \sim ae^{2x}, \; x \to \infty$

c)

$y(0) = 24$

$y'(0) = 16$

$y''(0) = 8$

$y'(x) = 2ae^{2x} - 2b\cos(2x) - 2c\sin(2x)$

$y''(x) = 4(ae^{2x} - b\sin(2x) - c\cos(2x))$

d)

$y(0) =$

$y'(0) =$

$y''(0) =$

c)

$y(0) = 0$

$y'(0) = 0$

$y''(0) = 0$

$m = 4(a - b\sin(0) - c\cos(0))$

$n = 2(a + b\cos(0) - c\sin(0))$

$p = a + b\sin(0) + c\cos(0)$

$a = 1$

$b = 0$

$c = 0$

$\Rightarrow$ $m = 4$, $n = 2$, $p = 1$, $a = 1$, $b = 0$, $c = 0$

d) $y(0) = 1$

$y'(0) = 2$

$y''(0) = 4$